Three-Dimensional Geological and Structural Modelling of a Shear Zone Using Leapfrog Geo

Integrated Drill Core Logging, Photogeology and Digital Geological Data Management

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Executive Summary

This project presents an integrated geological modelling workflow developed using Leapfrog Geo to construct a three-dimensional representation of lithological units and an interpreted shear zone from drillhole information. The study combines geological core logging, standardized drill core photography, photogeological interpretation, GIS data preparation, and implicit geological modelling to produce a structurally constrained subsurface model.

The workflow demonstrates how high-quality geological data can be transformed into a reliable three-dimensional interpretation suitable for mineral exploration and engineering decision-making. Additional emphasis is placed on digital geological data management inspired by Imago workflows and the potential application of Python automation to improve geological productivity, quality assurance, and reproducibility.

Chapter 1

Introduction

Three-dimensional geological modelling has become an essential component of modern mineral exploration because it enables geologists to visualize complex subsurface geological relationships that cannot be adequately interpreted from two-dimensional drill sections alone. Advances in implicit modelling software such as Leapfrog Geo have significantly improved the efficiency, consistency, and accuracy of geological interpretation by allowing geological observations to be integrated into dynamic three-dimensional models.

This project was undertaken to gain practical experience in drill core logging, geological interpretation, photogeology, digital geological data management, and three-dimensional implicit modelling using Leapfrog Geo. Throughout the project, geological observations obtained from drill cores were systematically logged, photographed, validated, and interpreted before being incorporated into an integrated geological model.

The project also explored the interpretation of a shear zone using geological evidence derived from lithological contacts, structural observations, and drillhole data. Guided structural modelling techniques were applied to define the orientation, continuity, and geometry of the interpreted shear zone.

Chapter 2

Geological Setting

Birimian Supergroup

The geological setting of Ghana is dominated by Paleoproterozoic Birimian rocks that form part of the West African Craton. These rocks consist primarily of metavolcanic sequences, metasedimentary rocks, granitoid intrusions, and volcanic belts that have undergone multiple phases of deformation and metamorphism.

The Birimian Supergroup hosts many of West Africa's largest gold deposits. Gold mineralization is strongly associated with regional deformation zones, particularly major shear zones that acted as pathways for hydrothermal fluids during mineralizing events.

These shear zones commonly exhibit:

- intense deformation
- foliated host rocks
- quartz veining
- alteration halos
- brecciation
- sulphide mineralization

Consequently, identifying and modelling shear zones is a fundamental aspect of exploration targeting in Birimian terrains.

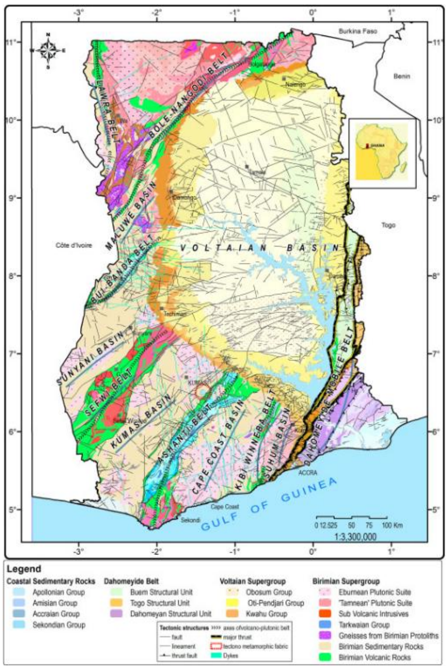

Figure 1 –(Geological Setting of the Birimian Supergroup)

Chapter 3

Project Objectives

The objectives of this study were to:

- validate drillhole datasets before modelling
- perform systematic geological core logging
- document drill cores using standardized photography
- interpret lithological and structural information
- prepare geological datasets for Leapfrog Geo
- construct three-dimensional lithological models
- identify and interpret an important shear zone
- generate a guided structural model
- demonstrate digital geological workflows that improve exploration efficiency

Chapter 4

Methodology

The project followed an integrated geological workflow consisting of thirteen major stages.

Step 1

Drillhole data validation

All collar, survey and lithological information were reviewed to ensure coordinate consistency, interval continuity and coding accuracy before modelling.

Step 2

Core tray preparation

Recovered drill cores were arranged sequentially and prepared for geological logging.

Step 3

Geological core logging

Lithological boundaries, alteration, mineralization, weathering, and structural observations were recorded following standard geological logging procedures.

Step 4

Drill core photography

A standardized imaging system using a Canon EOS 5D camera was employed to capture high-resolution photographs under controlled lighting conditions.

Step 5

Lithological interpretation

Lithological units were identified, coded and correlated across drillholes.

Step 6

Data validation

All geological intervals underwent quality assurance to ensure compatibility with Leapfrog Geo.

Step 7

Digital Elevation Model preparation

Terrain data were processed in QGIS to generate a digital elevation model for topographic control.

Step 8

Topography creation

The DEM was imported into Leapfrog Geo to create the project topographic surface.

Step 9

Drillhole import

Validated collar, survey and lithology tables were imported into Leapfrog Geo.

Step 10

Lithological modelling

Implicit interpolation was used to construct three-dimensional geological domains.

Step 11

Shear zone interpretation

Geological evidence from lithological relationships, structural observations and drillhole intersections was used to identify a continuous shear zone.

Step 12

Defining shear zone trend

Structural constraints were introduced to establish the regional orientation and continuity of the interpreted shear zone.

Step 13

Guided shear zone modelling

A guided implicit modelling workflow generated the final three-dimensional shear zone surface.

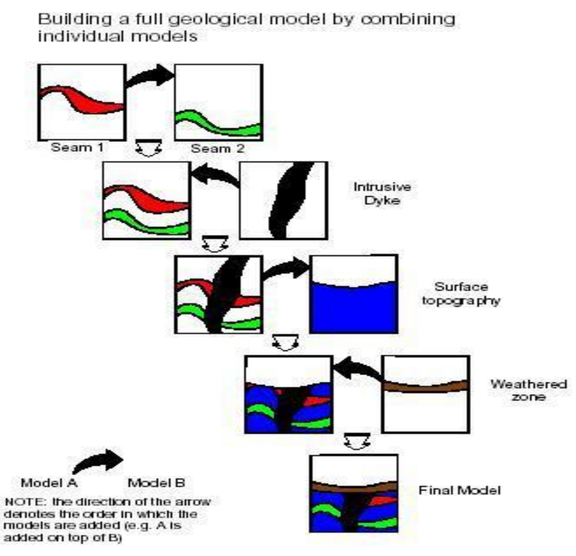

Figure 2- (Integrated Geological Workflow)

Chapter 5

Photogeology and Digital Core Management

Photogeology provides a permanent visual record of geological observations and significantly improves geological interpretation by allowing repeated examination of drill cores after physical logging has been completed.

Standardized drill core photography offers several advantages:

- permanent geological documentation
- improved quality assurance
- easier collaboration
- enhanced structural interpretation
- efficient geological review
- reduced transcription errors

Images were systematically organized according to drillhole, depth interval and project identifiers to ensure traceability throughout the modelling workflow.

Figure 3 –(Photogeology Workflow)

Chapter 6

Leapfrog Geo Modelling Workflow

Leapfrog Geo employs implicit modelling techniques that interpolate geological surfaces directly from geological observations rather than manually digitized wireframes.

The modelling process involved:

- importing validated drillholes
- generating project topography
- constructing lithological domains
- refining geological contacts
- introducing structural controls
- interpreting shear zone geometry
- generating an integrated geological model

The use of implicit modelling significantly reduced modelling time while maintaining geological consistency throughout the project.

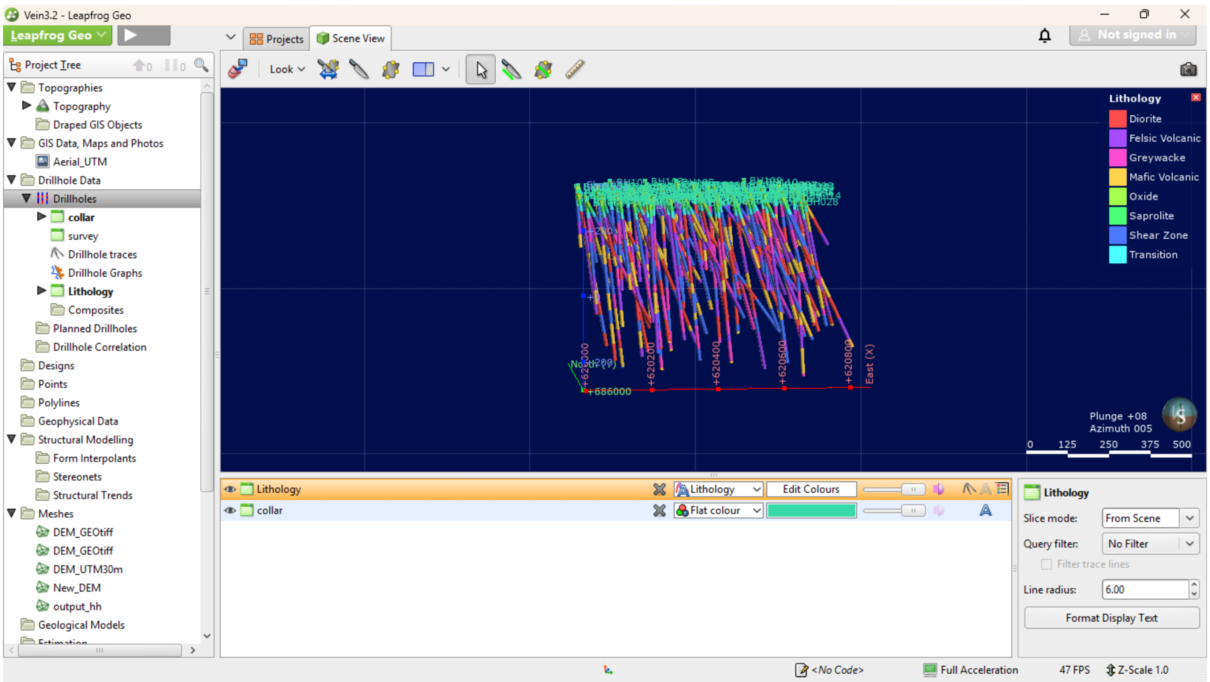

Figure 4 – (Leapfrog Geo Workflow)

Chapter 7

Shear Zone Interpretation

One of the major outcomes of this project was the interpretation and modelling of a structurally controlled shear zone.

The interpreted shear zone exhibits characteristics commonly associated with Birimian deformation corridors, including:

- consistent structural orientation
- continuity between drillholes
- lithological displacement
- deformation fabrics
- structural control of geological contacts

The interpreted trend was constrained using geological evidence rather than relying solely on mathematical interpolation, thereby improving geological realism.

The resulting guided shear zone model provides an effective structural framework for future exploration targeting and geological interpretation.

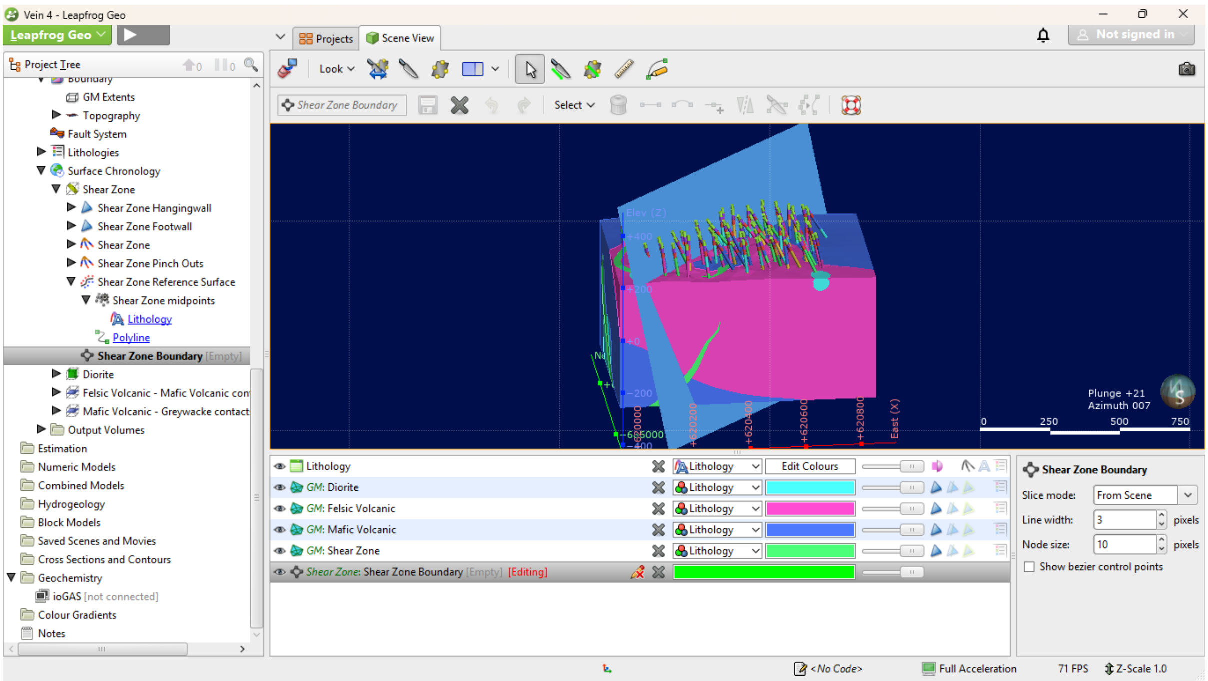

Figure 5 – (Shear Zone Interpretation)

Chapter 8

Python Automation and Imago-Inspired Digital Workflow

Although commercial platforms such as Imago provide sophisticated drill core management systems, similar workflows can be developed using Python to improve geological data management.

Python can automate:

- drillhole validation
- lithology consistency checking
- image renaming
- metadata generation
- batch image organization
- report generation
- geological statistics
- quality assurance
- data visualization

Such automation reduces repetitive manual tasks while improving data quality, consistency, and reproducibility.

The integration of Python with geological software creates opportunities for more efficient exploration workflows and supports modern digital mining practices.

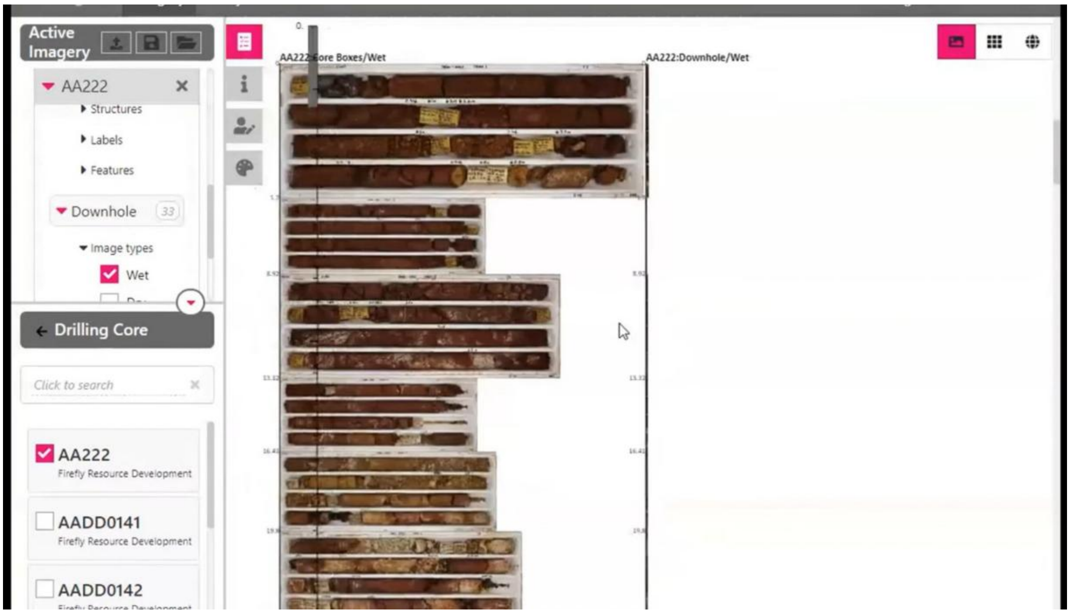

Figure 6 – (Python and Imago Workflow)

Chapter 9

Results

The project successfully achieved the following outcomes:

- validated geological database
- standardized drill core image library
- complete lithological interpretation
- three-dimensional geological model
- interpreted regional shear zone
- guided structural model
- improved understanding of geological relationships
- integrated digital geological workflow

Chapter 10

Discussion

This project demonstrates the importance of combining geological knowledge with digital modelling technologies. While Leapfrog Geo provides powerful interpolation algorithms, geological judgement remains essential for constructing meaningful geological models.

The inclusion of structural interpretation during modelling greatly improved the realism of the final model, particularly in defining the geometry of the interpreted shear zone. Similarly, standardized photogeology and digital data management enhanced traceability and confidence in geological observations.

The application of Python-based automation offers additional opportunities to streamline geological workflows by reducing manual processing and improving quality assurance.

Chapter 11

Conclusions

The integrated workflow successfully demonstrated the complete process of transforming raw geological observations into a three-dimensional geological model suitable for engineering and mineral exploration applications.

Through systematic drill core logging, standardized photography, photogeological interpretation, GIS preparation, implicit modelling and structural interpretation, the project produced a coherent geological model that accurately represents lithological relationships and the interpreted shear zone.

The project also highlights the growing importance of digital geological data management and automation. Integrating Python with geological software has the potential to significantly improve productivity, consistency and decision-making in future exploration projects.

References

Include references such as:

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Appendix

Figure 7- 03_core_logging_workflow

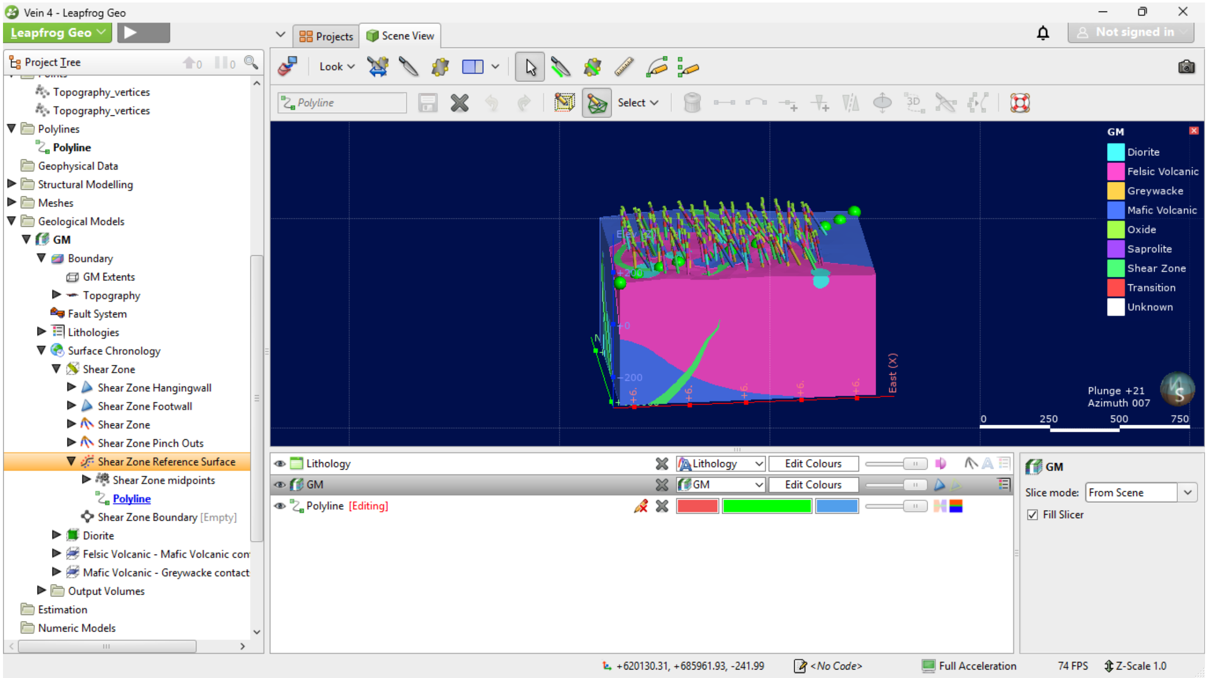

Figure 8- 08_final_model with drillholes, without erosional surface